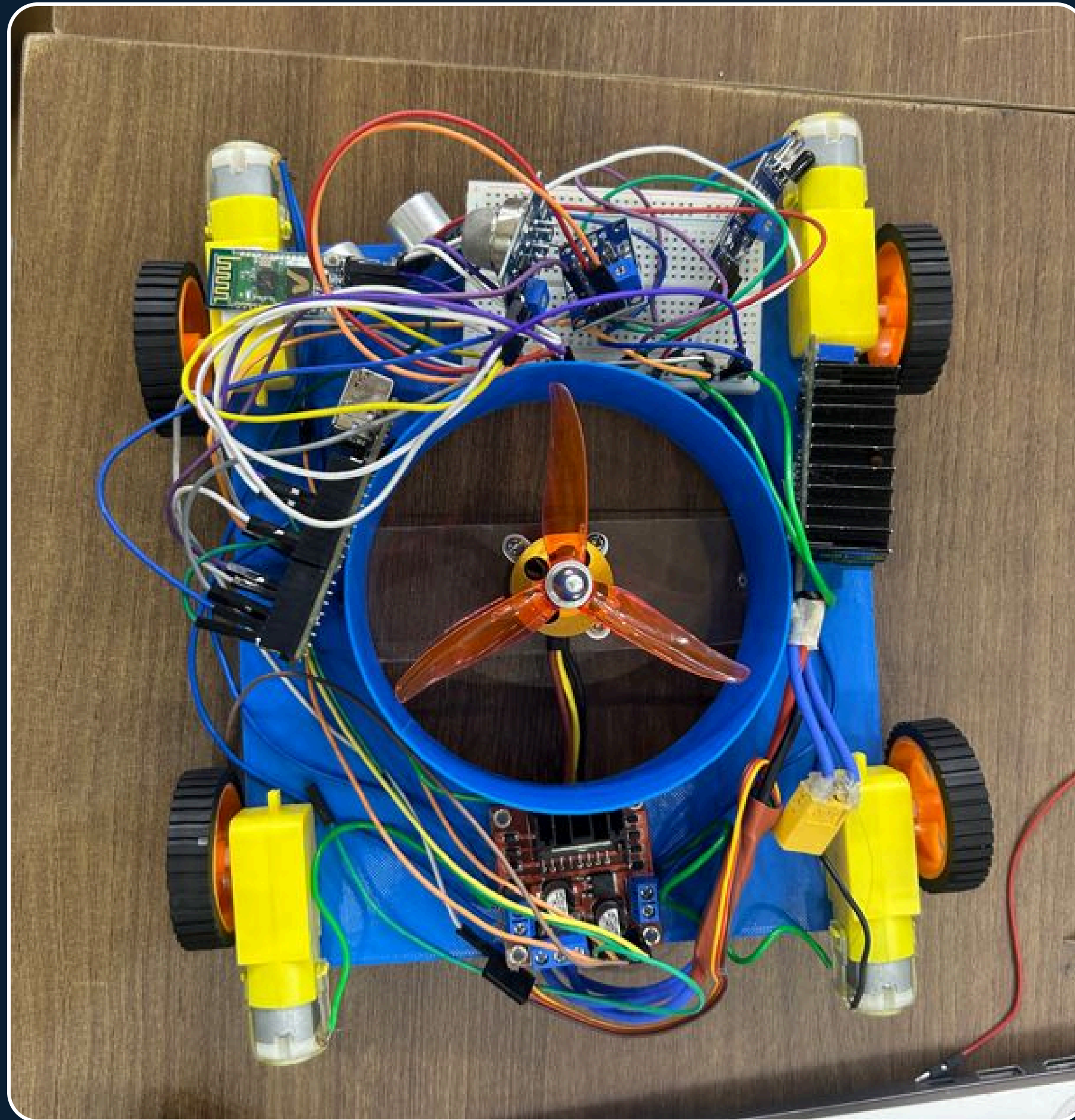




404 ERROR SOLVED

WALL CLIMBING INSPECTION ROBOT

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Problem Statement:

- Risky Manual Inspection : Inspection of high-rise buildings, bridges, storage tanks, and industrial structures is typically performed manually using scaffolding, ropes, or cranes. These methods expose workers to significant safety risks while operating at dangerous heights.
- High Cost and Time Consumption : Traditional inspection techniques require specialized equipment, skilled personnel, and extensive setup time, leading to increased operational costs and inefficient inspection processes.
- Limited Accessibility and Inspection Accuracy : Many structural surfaces are difficult to access, making thorough inspection challenging. Manual inspection may fail to detect small defects such as cracks, corrosion, or surface deterioration at early stages.
- Need for a Safer and Smarter Solution : There is a need for an automated or robotic system capable of safely accessing vertical surfaces and performing efficient, real-time inspection with minimal human risk.



GENESIS

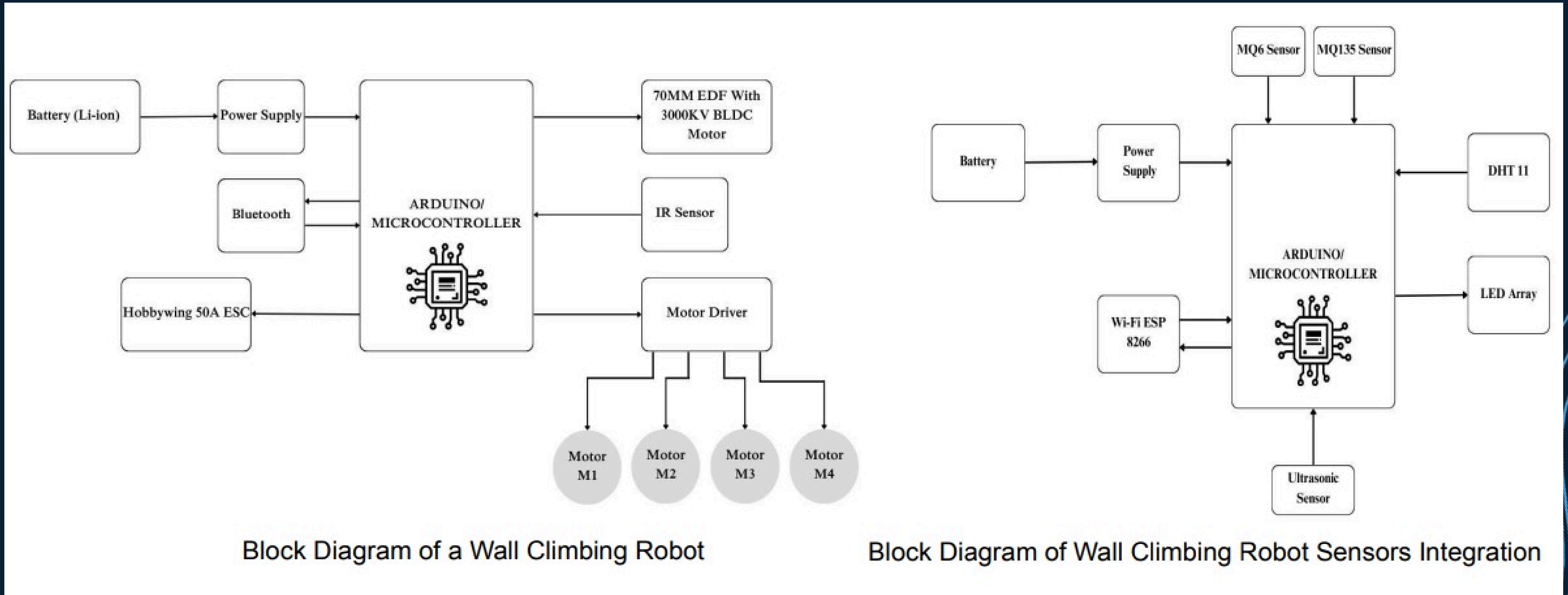
Proposed Idea/ Solution

PROPOSED SOLUTION	HOW IT ADDRESSES THE PROBLEM	INNOVATION AND UNIQUENESS
A small robot that sticks to walls and crawls safely.	Keeps humans away from risky heights (no ropes/scaffolding).	Uses TinyML to detect the onboard crack and will send alert to the user
Records and monitors the surface in real time.Detects cracks, rust, and leaks during inspection.	Reaches hard-to-access areas quickly (tall buildings, tanks, bridges).	Unique wall-sticking and crawling mechanism where drones can't work well.
Shares a live view with the operator.	Provides consistent checks with digital records for comparison.	Combines visual inspection along with surface sensing in one pass.
Follows a set path for full surface coverage.	Saves time and cost compared to manual inspection.	Creates a defect map/report from raw data.
Built-in safety features: stops near edges and has a tether to prevent falls	Can also be used for crack detection in vehicles such as cars, airplanes, etc and also in nuclear power plants.	Can adapt to different surfaces (steel, concrete, glass).

How it Works(Technical Approach)

- Adhesion Mechanism: Uses a EDF based vacuum suction system that creates a pressure differential, allowing the robot to carry a payload (sensors/cameras) without falling.
- Hardware Stack: Control Unit: ESP32 & ESP32-CAM for high-level image processing along with a synchronised transmitter - receiver to control the bot
- Sensing Layer: Ultrasonic sensors (obstacle avoidance), DHT11 (ambient conditions), and MQ sensors (gas leak detection in industrial tanks).
- TinyML Integration: Onboard AI models trained using Edge Impulse to classify "Crack" vs. "No Crack" in real-time, reducing the bandwidth needed for data transmission.
- Software: Python-based GUI for the operator and Arduino C++ for hardware logic.

How it Works(Technical Approach)



Benefits of Solution

- Edge Intelligence: Unlike standard "dumb" crawlers, our robot processes data on the device (TinyML). It doesn't just record; it analyzes.
- Multi-Sensor Fusion: Combines visual data with environmental data (gas, temp, humidity) to provide a holistic "Health Score" of the structure.
- Drone Alternative: Drones have limited battery life (20-30 mins) and struggle with wind gusts near buildings. Our robot is physically connected to the surface, making it more stable and energy-efficient for long-duration scans.
- Cost-Effective Design: Built using off-the-shelf high-performance components, making it accessible for MSMEs (Small/Medium Enterprises), not just giant corporations.



Innovation & Uniqueness

- Industrial Use: Inspection of oil & gas storage tanks and chemical reactors without human entry (eliminating toxic exposure).
- Infrastructure: Regular health checks for bridges and railway overpasses to prevent collapses.
- Renewable Energy: Cleaning and inspecting solar panels on tilted rooftops or high-rise glass facades.
- Defense & Rescue: Surveillance in urban combat or checking for structural integrity in earthquake-hit buildings before rescue teams enter.
- Market Potential: Directly addresses the multi-billion dollar NDT (Non-Destructive Testing) market.

Impact & Applications

- Enhanced Safety: Removes human risk by deploying robots in hazardous inspection zones.
- Improved Efficiency: Conducts faster, frequent, and automated inspections, minimizing downtime.
- Cost-Effective: Cuts expenses on scaffolding, cranes, and manual labor.
- High-Quality Insights: Delivers real-time video, crack classification via TinyML, and environmental sensing for accurate decisions.
- Versatility: Adaptable for infrastructure monitoring, solar panel/window cleaning, defense, and disaster response.
- Sustainability: Enables early fault detection, lowering repair costs and preventing material wastage.
- Applications: Oil & Gas tanks, Wind Turbines, Nuclear Cooling Towers, and Bridge Pylons.
- Key Benefit: Eliminates human risk, ensures 100% inspection coverage, and provides digital records for predictive maintenance.

